
BEU
3nd Semester cse Branch
OPERATING
SYSTEM (OS)
MCQ QUESTION
EDUCATION
PLUS+

Unit 1

Introduction

Questions 1 - 18

Q1 U1: Introduction

Which of the following is NOT a type of Operating System?

- **A** Real-time Operating System
- **B** Batch Operating System
- **C** Hybrid Operating System
- **D** Time-sharing Operating System

Think about your answer...

Q1 ANSWER

Which of the following is NOT a type of Operating System?

A Real-time Operating System

B Batch Operating System

C **Hybrid Operating System** ✓

D Time-sharing Operating System

EXPLANATION

A Hybrid Operating System is not a standard classification. The main types are Batch, Time-sharing, Real-time, Distributed, and Network OS.

Q2 U1: Introduction

The primary purpose of an operating system is to:

- **A** Provide a user interface
- **B** Manage hardware resources and provide an abstraction layer
- **C** Compile programs
- **D** Design algorithms

Think about your answer...

Q2 **ANSWER**

The primary purpose of an operating system is to:

- **A** Provide a user interface
- **B** **Manage hardware resources and provide an abstraction layer** ✓
- **C** Compile programs
- **D** Design algorithms

EXPLANATION

The OS acts as an intermediary between hardware and user applications, managing resources like CPU, memory, and I/O devices while providing abstraction.

Q3

U1: Introduction

Which generation of operating systems introduced multiprogramming?

A First Generation

B Second Generation

C Third Generation

D Fourth Generation

Think about your answer...

Q3

ANSWER

Which generation of operating systems introduced multiprogramming?

A First Generation

B Second Generation

C **Third Generation** ✓

D Fourth Generation

EXPLANATION

Multiprogramming was introduced in the Third Generation (mid-1960s) with systems like IBM OS/360, allowing multiple programs in memory simultaneously.

Q4 U1: Introduction

A system call is:

A A function call within a user program

B An interface between a process and the operating system

C A hardware interrupt

D A type of scheduling algorithm

Think about your answer...

Q4 ANSWER

A system call is:

A A function call within a user program

B **An interface between a process and the operating system** ✓

C A hardware interrupt

D A type of scheduling algorithm

EXPLANATION

A system call provides the interface between a running process and the operating system, allowing user programs to request OS services.

Q5

In a monolithic kernel architecture:

- **A** All OS services run in user space
- **B** All OS services run in kernel space
- **C** Only file management runs in kernel space
- **D** Device drivers run in user space

Think about your answer...

Q5

ANSWER

In a monolithic kernel architecture:

A All OS services run in user space

B **All OS services run in kernel space** ✓

C Only file management runs in kernel space

D Device drivers run in user space

EXPLANATION

In a monolithic kernel, all OS services (file system, device drivers, etc.) run in kernel space as a single large program.

Q6

U1: Introduction

Which OS structure provides the best modularity and extensibility?

- **A** Monolithic
- **B** Layered
- **C** Microkernel
- **D** Simple

Think about your answer...

Q6 ANSWER

Which OS structure provides the best modularity and extensibility?

A Monolithic

B Layered

C **Microkernel** ✓

D Simple

EXPLANATION

The microkernel approach provides the best modularity by moving most services to user space, keeping only essential functions in the kernel.

Q7

U1: Introduction

The concept of Virtual Machine was first introduced by:

A IBM

B Microsoft

C Apple

D Google

Think about your answer...

Q7 **ANSWER**

The concept of Virtual Machine was first introduced by:

- **A IBM** ✓
- B Microsoft
- C Apple
- D Google

EXPLANATION

The concept of Virtual Machines was first introduced by IBM with CP-40 and CP-67 in the 1960s.

Q8

In the layered approach to OS design, each layer is built upon:

- **A** The hardware only
- **B** The layer below it
- **C** The layer above it
- **D** All other layers

Think about your answer...

Q8 ANSWER

In the layered approach to OS design, each layer is built upon:

A The hardware only

B **The layer below it** ✓

C The layer above it

D All other layers

EXPLANATION

In the layered approach, each layer is built upon the services and functions provided by the lower layers, creating a hierarchical structure.

Q9

U1: Introduction

Which of the following is an advantage of the microkernel approach?

- **A** Higher performance
- **B** Easier to extend and port
- **C** Less communication overhead
- **D** Simpler implementation

Think about your answer...

Q9

ANSWER

Which of the following is an advantage of the microkernel approach?

A Higher performance

B **Easier to extend and port** ✓

C Less communication overhead

D Simpler implementation

EXPLANATION

The microkernel approach offers better extensibility since most services run in user space and can be added or modified without changing the kernel.

Q10 U1: Introduction

UNIX operating system is an example of:

- **A** Microkernel-based OS
- **B** Monolithic OS
- **C** Layered OS
- **D** Client-Server OS

Think about your answer...

Q10 ANSWER

UNIX operating system is an example of:

- A Microkernel-based OS
- B **Monolithic OS** ✓
- C Layered OS
- D Client-Server OS

EXPLANATION

UNIX uses a monolithic kernel architecture where most OS services run in kernel space, though it supports loadable kernel modules.

The first operating system was developed for which generation of computers?

A First Generation (vacuum tubes)

B Second Generation (transistors)

C Third Generation (ICs)

D Fourth Generation (VLSI)

Think about your answer...

Q11 ANSWER

The first operating system was developed for which generation of computers?

- **A First Generation (vacuum tubes) ✓**
- B Second Generation (transistors)
- C Third Generation (ICs)
- D Fourth Generation (VLSI)

EXPLANATION

The first operating systems were developed for First Generation computers (1940s-1950s) to manage basic I/O and job sequencing.

Q12

Which of the following is NOT an OS service?

- A Program execution
- B I/O operations
- C File system manipulation
- D Hardware manufacturing

Think about your answer...

Q12

ANSWER

Which of the following is NOT an OS service?

A Program execution

B I/O operations

C File system manipulation

D **Hardware manufacturing** ✓

EXPLANATION

Compiling programs is a function of the compiler, not an OS service. OS services include program execution, I/O operations, and file management.

Q13

U1: Introduction

Windows Operating System uses which kernel architecture?

- **A** Pure Microkernel
- **B** Hybrid Kernel
- **C** Exokernel
- **D** Nanokernel

Think about your answer...

Q13 ANSWER

Windows Operating System uses which kernel architecture?

A Pure Microkernel

B Hybrid Kernel ✓

C Exokernel

D Nanokernel

EXPLANATION

Windows uses a hybrid kernel architecture combining elements of both monolithic and microkernel designs.

Q14

U1: Introduction

The main disadvantage of the layered OS structure is:

- A Poor modularity
- B Difficulty in defining layer boundaries
- C Inability to support multiple users
- D Lack of security

Think about your answer...

Q14

ANSWER

The main disadvantage of the layered OS structure is:

A Poor modularity

B Difficulty in defining layer boundaries ✓

C Inability to support multiple users

D Lack of security

EXPLANATION

The main disadvantage is that it is difficult to define and maintain strict layer boundaries, and layers may need to call functions across layers.

Q15

U1: Introduction

A virtual machine provides:

- **A** A physical machine
- **B** An isolated duplicate of a real machine
- **C** A new hardware platform
- **D** A network connection

Think about your answer...

Q15

ANSWER

A virtual machine provides:

A A physical machine

B An isolated duplicate of a real machine ✓

C A new hardware platform

D A network connection

EXPLANATION

A virtual machine provides an isolated execution environment that behaves like a separate physical computer with its own OS and resources.

Q16 U1: Introduction

Which system call is used to create a new process in UNIX?

A exec()

B fork()

C wait()

D exit()

Think about your answer...

Q16 ANSWER

Which system call is used to create a new process in UNIX?

A exec()

B **fork()** ✓

C wait()

D exit()

EXPLANATION

The fork() system call is used in UNIX to create a new process by duplicating the calling process.

Q17

U1: Introduction

Batch processing operating systems were used in which generation?

A First Generation

B Second Generation

C Third Generation

D Fourth Generation

Think about your answer...

Q17

ANSWER

Batch processing operating systems were used in which generation?

A First Generation

B **Second Generation** ✓

C Third Generation

D Fourth Generation

EXPLANATION

Batch processing operating systems were used in the Second Generation of computers (late 1950s to mid-1960s).

Q18

U1: Introduction

The primary difference between user mode and kernel mode is:

A Speed of execution

B Access to privileged instructions

C Memory allocation

D File permissions

Think about your answer...

Q18 ANSWER

The primary difference between user mode and kernel mode is:

- **A** Speed of execution
- **B** **Access to privileged instructions** ✓
- **C** Memory allocation
- **D** File permissions

EXPLANATION

The primary difference is the level of privilege: kernel mode has full hardware access, while user mode has restricted access to prevent system damage.

Unit 2

Processes

Questions 19 - 38

Q19

U2: Processes

A process in the ready state is:

- **A** Waiting for I/O
- **B** Waiting for CPU allocation
- **C** Being executed by the CPU
- **D** Terminated

Think about your answer...

Q19 ANSWER

A process in the ready state is:

A Waiting for I/O

B Waiting for CPU allocation ✓

C Being executed by the CPU

D Terminated

EXPLANATION

A process in the ready state is waiting to be assigned to a processor; it has all resources except the CPU.

Q20

U2: Processes

The Process Control Block (PCB) contains:

- **A** Only the process ID
- **B** Process state, program counter, CPU registers, and more
- **C** Only memory allocation details
- **D** Only scheduling information

Think about your answer...

Q20 ANSWER

The Process Control Block (PCB) contains:

A Only the process ID

B **Process state, program counter, CPU registers, and more** ✓

C Only memory allocation details

D Only scheduling information

EXPLANATION

The PCB contains process state, program counter, CPU registers, memory limits, and accounting information for process management.

Q21

U2: Processes

Context switching involves:

- **A** Saving the state of the old process and loading the saved state of the new process
- **B** Deleting the old process
- **C** Creating a new process
- **D** Terminating the running process

Think about your answer...

Q21 ANSWER

Context switching involves:

A Saving the state of the old process and loading the saved state of the new process ✓

B Deleting the old process

C Creating a new process

D Terminating the running process

EXPLANATION

Context switching involves saving the state of the currently running process and loading the saved state of the new process to switch CPU allocation.

Q22

U2: Processes

A thread is also known as:

- **A** A heavyweight process
- **B** A lightweight process
- **C** A virtual process
- **D** A kernel process

Think about your answer...

Q22 ANSWER

A thread is also known as:

A. A heavyweight process

B. **A lightweight process** ✓

C. A virtual process

D. A kernel process

EXPLANATION

A thread is also known as a lightweight process because it shares the process's resources and has less overhead than a full process.

Q23

U2: Processes

User-level threads are managed by:

- **A** The operating system kernel
- **B** The user-level library
- **C** The hardware
- **D** The compiler

Think about your answer...

Q23 ANSWER

User-level threads are managed by:

A The operating system kernel

B **The user-level library** ✓

C The hardware

D The compiler

EXPLANATION

User-level threads are managed by a thread library in user space, without kernel involvement, making thread operations faster.

Q24

U2: Processes

Kernel-level threads are managed by:

A User library

B Application program

C Operating system kernel

D Hardware controller

Think about your answer...

Q24 ANSWER

Kernel-level threads are managed by:

A User library

B Application program

C **Operating system kernel** ✓

D Hardware controller

EXPLANATION

Kernel-level threads are managed directly by the operating system kernel, which handles scheduling and context switching.

Q25 U2: Processes

In the FCFS scheduling algorithm, the process that arrives first:

- **A** Gets the least CPU time
- **B** Is executed first
- **C** Is executed last
- **D** Gets highest priority

Think about your answer...

Q25 ANSWER

In the FCFS scheduling algorithm, the process that arrives first:

A Gets the least CPU time

B **Is executed first** ✓

C Is executed last

D Gets highest priority

EXPLANATION

In FCFS, the process that arrives first gets the CPU first; it is a simple, non-preemptive scheduling algorithm.

Q26

U2: Processes

The SJF scheduling algorithm may cause:

A Aging

B Starvation of long processes

C Deadlock

D Priority inversion

Think about your answer...

Q26 ANSWER

The SJF scheduling algorithm may cause:

A Aging

B Starvation of long processes ✓

C Deadlock

D Priority inversion

EXPLANATION

SJF may cause starvation (indefinite blocking) of longer processes as shorter processes keep arriving and getting priority.

Q27

U2: Processes

Round Robin scheduling uses:

A Priority queue

B Time quantum

C Shortest job first

D First come first served

Think about your answer...

Q27

ANSWER

Round Robin scheduling uses:

A Priority queue

B Time quantum ✓

C Shortest job first

D First come first served

EXPLANATION

Round Robin uses a time quantum (time slice) to allocate CPU time equally among processes in a circular order.

Q28

U2: Processes

Which scheduling algorithm is preemptive?

- A FCFS
- B SJF (non-preemptive)
- C Round Robin
- D SJF (cooperative)

Think about your answer...

Q28 ANSWER

Which scheduling algorithm is preemptive?

A FCFS

B SJF (non-preemptive)

C **Round Robin** ✓

D SJF (cooperative)

EXPLANATION

SRTF (Shortest Remaining Time First) is the preemptive version of SJF, where a new shorter process can preempt the running one.

Q29

U2: Processes

Turnaround time is defined as:

- **A** Time spent in the ready queue
- **B** Total time from submission to completion
- **C** Time spent executing on CPU
- **D** Time waiting for I/O

Think about your answer...

Q29 ANSWER

Turnaround time is defined as:

- **A** Time spent in the ready queue
- **B** **Total time from submission to completion** ✓
- **C** Time spent executing on CPU
- **D** Time waiting for I/O

EXPLANATION

Turnaround time is the total time from process submission to completion, including both waiting and execution time.

Q30

U2: Processes

CPU utilization is maximized when:

- **A** Processes are I/O-bound
- **B** Processes are CPU-bound
- **C** There is a mix of CPU-bound and I/O-bound processes
- **D** Only one process runs

Think about your answer...

Q30 ANSWER

CPU utilization is maximized when:

A Processes are I/O-bound

B Processes are CPU-bound

C There is a mix of CPU-bound and I/O-bound processes ✓

D Only one process runs

EXPLANATION

CPU utilization is maximized when the degree of multiprogramming is high, keeping the CPU busy with always having a process ready to run.

Q31

U2: Processes

In the Rate Monotonic (RM) scheduling algorithm:

A. Tasks with shorter periods get higher priority

B. Tasks with longer periods get higher priority

C. Priority is assigned based on execution time

D. Priority is assigned randomly

Think about your answer...

Q31 ANSWER

In the Rate Monotonic (RM) scheduling algorithm:

A **Tasks with shorter periods get higher priority** ✓

B Tasks with longer periods get higher priority

C Priority is assigned based on execution time

D Priority is assigned randomly

EXPLANATION

In Rate Monotonic scheduling, tasks with shorter periods (higher frequency) are assigned higher priority for optimal static-priority scheduling.

Q32 U2: Processes

The Earliest Deadline First (EDF) scheduling algorithm is:

- **A** Static priority-based
- **B** Dynamic priority-based
- **C** Non-preemptive only
- **D** Based on arrival time

Think about your answer...

Q32 ANSWER

The Earliest Deadline First (EDF) scheduling algorithm is:

- A Static priority-based
- **B Dynamic priority-based** ✓
- C Non-preemptive only
- D Based on arrival time

EXPLANATION

EDF is a dynamic scheduling algorithm where the task with the earliest deadline is assigned the highest priority.

Q33

U2: Processes

The benefit of multithreading is:

A Increased memory usage

B Responsiveness and resource sharing

C Slower execution

D More context switching overhead

Think about your answer...

Q33 **ANSWER**

The benefit of multithreading is:

A Increased memory usage

B Responsiveness and resource sharing ✓

C Slower execution

D More context switching overhead

EXPLANATION

Multithreading benefits include improved responsiveness, resource sharing, economy (cheaper than multiprocessing), and utilization of multiprocessor architectures.

Q34

U2: Processes

Which of the following is NOT a process state?

A Ready

B Running

C Compiling

D Waiting

Think about your answer...

Q34 ANSWER

Which of the following is NOT a process state?

A Ready

B Running

C **Compiling** ✓

D Waiting

EXPLANATION

'Terminated' is a process state, not excluded from process states. The options that are NOT states would depend on the list, but 'Compiled' is not a standard process state.

Q35

U2: Processes

Throughput refers to:

- **A** Time taken by a process to complete
- **B** Number of processes completed per unit time
- **C** Time spent waiting in ready queue
- **D** CPU speed

Think about your answer...

Q35

ANSWER

Throughput refers to:

A Time taken by a process to complete

B **Number of processes completed per unit time** ✓

C Time spent waiting in ready queue

D CPU speed

EXPLANATION

Throughput refers to the number of processes completed per unit of time, measuring system productivity.

Q36

U2: Processes

The long-term scheduler selects processes from:

A Ready queue to running

B Job queue to ready queue

C Running to waiting

D Waiting to ready

Think about your answer...

Q36 ANSWER

The long-term scheduler selects processes from:

A Ready queue to running

B Job queue to ready queue ✓

C Running to waiting

D Waiting to ready

EXPLANATION

The long-term scheduler selects processes from the job pool and loads them into memory for execution.

Q37

U2: Processes

In multiprocessor scheduling, load sharing means:

- **A** Only one processor handles all processes
- **B** Processors share the common ready queue
- **C** Each processor has its own private queue
- **D** Processes are allocated fixed processors

Think about your answer...

Q37

ANSWER

In multiprocessor scheduling, load sharing means:

A Only one processor handles all processes

B Processors share the common ready queue ✓

C Each processor has its own private queue

D Processes are allocated fixed processors

EXPLANATION

Load sharing means that processes are distributed across multiple processors so that no processor is idle while others have work.

Q38

U2: Processes

Which metric measures the time a process spends waiting in the ready queue?

A Turnaround time

B Response time

C Waiting time

D Throughput

Think about your answer...

Q38 ANSWER

Which metric measures the time a process spends waiting in the ready queue?

A Turnaround time

B Response time

C Waiting time ✓

D Throughput

EXPLANATION

Waiting time is the metric that measures the time a process spends waiting in the ready queue before getting CPU time.

Unit 3

Inter-process Communication

Questions 39 - 56

Q39

U3: Inter-process Communication

A critical section is a segment of code where:

A Processes can execute freely

B Shared resources are accessed

C No process can enter

D Only the kernel can execute

Think about your answer...

Q39

ANSWER

A critical section is a segment of code where:

A Processes can execute freely

B **Shared resources are accessed** ✓

C No process can enter

D Only the kernel can execute

EXPLANATION

A critical section is a segment of code where a process accesses shared resources that must not be concurrently accessed by other processes.

Q40

U3: Inter-process Communication

A race condition occurs when:

- **A** A process is terminated abruptly
- **B** The outcome depends on the order of execution of processes
- **C** A process is in deadlock
- **D** CPU is idle

Think about your answer...

Q40 ANSWER

A race condition occurs when:

- **A** A process is terminated abruptly
- **B** **The outcome depends on the order of execution of processes** ✓
- **C** A process is in deadlock
- **D** CPU is idle

EXPLANATION

A race condition occurs when the outcome of process execution depends on the order in which processes are scheduled and executed.

Q41

U3: Inter-process Communication

Mutual exclusion means:

- **A** All processes can access the critical section simultaneously
- **B** Only one process can be in its critical section at a time
- **C** No process can ever enter the critical section
- **D** Processes must share all variables

Think about your answer...

Q41 ANSWER

Mutual exclusion means:

A All processes can access the critical section simultaneously

B Only one process can be in its critical section at a time ✓

C No process can ever enter the critical section

D Processes must share all variables

EXPLANATION

Mutual exclusion means that at most one process can be in its critical section at any given time, preventing concurrent access to shared resources.

Q42

U3: Inter-process Communication

Peterson's solution for mutual exclusion works for:

A Any number of processes

B Two processes

C Three processes

D Unlimited processes

Think about your answer...

Q42

ANSWER

Peterson's solution for mutual exclusion works for:

A Any number of processes

B **Two processes** ✓

C Three processes

D Unlimited processes

EXPLANATION

Peterson's solution works for two processes, using shared variables (turn and flag) to ensure mutual exclusion.

Q43

U3: Inter-process Communication

A semaphore with value 1 is called:

- **A** Counting semaphore
- **B** Binary semaphore
- **C** Mutex
- **D** Both B and C

Think about your answer...

Q43 ANSWER

A semaphore with value 1 is called:

A Counting semaphore

B Binary semaphore

C Mutex

D **Both B and C** ✓

EXPLANATION

A semaphore with value 1 is called a binary semaphore (or mutex), which can only take values 0 and 1.

Q44

U3: Inter-process Communication

The wait (P) operation on a semaphore:

- A Increments the semaphore value
- B Decrements the semaphore value
- C Does not change the value
- D Doubles the value

Think about your answer...

Q44

ANSWER

The wait (P) operation on a semaphore:

A Increments the semaphore value

B **Decrements the semaphore value** ✓

C Does not change the value

D Doubles the value

EXPLANATION

The wait (P) operation decrements the semaphore value; if the value becomes negative, the process is blocked.

The signal (V) operation on a semaphore:

- **A** Decrements the semaphore value
- **B** Increments the semaphore value
- **C** Does not change the value
- **D** Resets the value to zero

Think about your answer...

Q45 ANSWER

The signal (V) operation on a semaphore:

A Decrements the semaphore value

B Increments the semaphore value ✓

C Does not change the value

D Resets the value to zero

EXPLANATION

The signal (V) operation increments the semaphore value; if processes are waiting, one is awakened.

Q46

U3: Inter-process Communication

In the Producer-Consumer problem, the producer cannot add items when:

- **A** The buffer is empty
- **B** The buffer is full
- **C** The consumer is running
- **D** The buffer has one item

Think about your answer...

Q46 ANSWER

In the Producer-Consumer problem, the producer cannot add items when:

A The buffer is empty

B The buffer is full ✓

C The consumer is running

D The buffer has one item

EXPLANATION

In the Producer-Consumer problem, the producer cannot add items when the buffer is full (semaphore count is zero).

A monitor is a high-level synchronization construct that:

- **A** Uses only busy waiting
- **B** Provides condition variables for synchronization
- **C** Cannot be used for mutual exclusion
- **D** Requires hardware support

Think about your answer...

Q47 ANSWER

A monitor is a high-level synchronization construct that:

A Uses only busy waiting

B **Provides condition variables for synchronization** ✓

C Cannot be used for mutual exclusion

D Requires hardware support

EXPLANATION

A monitor is a high-level synchronization construct that allows only one process to be active within it at a time, using condition variables for coordination.

Q48

U3: Inter-process Communication

In the Dining Philosopher problem, a deadlock can occur when:

- **A** All philosophers pick up their left fork simultaneously
- **B** One philosopher eats
- **C** Forks are unlimited
- **D** All philosophers pick up their right fork simultaneously

Think about your answer...

Q48 ANSWER

In the Dining Philosopher problem, a deadlock can occur when:

- **A** **All philosophers pick up their left fork simultaneously** ✓
- **B** One philosopher eats
- **C** Forks are unlimited
- **D** All philosophers pick up their right fork simultaneously

EXPLANATION

In the Dining Philosopher problem, deadlock can occur when each philosopher picks up their left fork simultaneously, creating a circular wait.

Q49

U3: Inter-process Communication

In the Readers-Writers problem, which is allowed?

A. Multiple readers and one writer simultaneously

B. Multiple readers reading simultaneously

C. Multiple writers writing simultaneously

D. One reader and one writer simultaneously

Think about your answer...

Q49 ANSWER

In the Readers-Writers problem, which is allowed?

A Multiple readers and one writer simultaneously

B **Multiple readers reading simultaneously** ✓

C Multiple writers writing simultaneously

D One reader and one writer simultaneously

EXPLANATION

In the Readers-Writers problem, multiple readers can read simultaneously, but writers need exclusive access.

Q50

U3: Inter-process Communication

Message passing is suitable for:

A Only single-processor systems

B Distributed systems

C Only small data transfers

D Only kernel-level communication

Think about your answer...

Q50 ANSWER

Message passing is suitable for:

- **A** Only single-processor systems
- **B** **Distributed systems** ✓
- **C** Only small data transfers
- **D** Only kernel-level communication

EXPLANATION

Message passing is suitable for distributed systems where processes don't share memory and must communicate through messages.

Q51 U3: Inter-process Communication

Shared memory IPC requires:

- **A** No synchronization
- **B** Synchronization mechanisms to avoid race conditions
- **C** Only one process
- **D** Hardware interrupts

Think about your answer...

Q51 ANSWER

Shared memory IPC requires:

A No synchronization

B Synchronization mechanisms to avoid race conditions ✓

C Only one process

D Hardware interrupts

EXPLANATION

Shared memory IPC requires processes to explicitly handle synchronization and mutual exclusion, as the OS doesn't provide it for shared memory regions.

Q52

U3: Inter-process Communication

Strict alternation is a solution for mutual exclusion that:

- **A** Allows both processes to enter critical section simultaneously
- **B** Requires processes to take turns entering critical section
- **C** Does not guarantee mutual exclusion
- **D** Is only for reader-writer problems

Think about your answer...

Q52 ANSWER

Strict alternation is a solution for mutual exclusion that:

A Allows both processes to enter critical section simultaneously

B Requires processes to take turns entering critical section ✓

C Does not guarantee mutual exclusion

D Is only for reader-writer problems

EXPLANATION

Strict alternation forces processes to take turns entering critical sections, but it doesn't satisfy the progress requirement since a process not in its critical section can block others.

Q53

U3: Inter-process Communication

Hardware solutions to mutual exclusion use:

A. Software flags only

B. Special machine instructions like Test-and-Set

C. Semaphores

D. Monitors

Think about your answer...

Q53 ANSWER

Hardware solutions to mutual exclusion use:

A Software flags only

B Special machine instructions like Test-and-Set ✓

C Semaphores

D Monitors

EXPLANATION

Hardware solutions use special machine instructions like Test-and-Set and Swap that are atomic and cannot be interrupted.

Q54

U3: Inter-process Communication

The disadvantage of busy waiting (spinlock) is:

- A It wastes CPU cycles
- B It is too slow to respond
- C It causes deadlock
- D It does not ensure mutual exclusion

Think about your answer...

Q54 ANSWER

The disadvantage of busy waiting (spinlock) is:

- **A It wastes CPU cycles** ✓
- B It is too slow to respond
- C It causes deadlock
- D It does not ensure mutual exclusion

EXPLANATION

The disadvantage of busy waiting (spinlock) is that it wastes CPU cycles while a process continuously checks for a condition to become true.

Q55

U3: Inter-process Communication

An event counter is used to:

- **A** Count the number of instructions executed
- **B** Synchronize processes by counting occurrences of an event
- **C** Measure CPU time
- **D** Track memory usage

Think about your answer...

Q55 ANSWER

An event counter is used to:

A Count the number of instructions executed

B **Synchronize processes by counting occurrences of an event** ✓

C Measure CPU time

D Track memory usage

EXPLANATION

An event counter is used to synchronize processes by tracking the number of times an event has occurred.

Q56

U3: Inter-process Communication

Direct communication in message passing requires:

- A: A shared mailbox
- B: Processes to name each other explicitly
- C: Indirect addressing
- D: Kernel intervention

Think about your answer...

Q56

ANSWER

Direct communication in message passing requires:

A. A shared mailbox

B. **Processes to name each other explicitly** ✓

C. Indirect addressing

D. Kernel intervention

EXPLANATION

Direct communication requires processes to explicitly name each other (the sender and receiver must identify each other).

Unit 4

Deadlocks

Questions 57 - 71

Q57

U4: Deadlocks

Which of the following is a necessary condition for deadlock?

- A Preemption
- B Mutual exclusion
- C Starvation
- D Aging

Think about your answer...

Q57 ANSWER

Which of the following is a necessary condition for deadlock?

A Preemption

B **Mutual exclusion** ✓

C Starvation

D Aging

EXPLANATION

Mutual exclusion is one of the four necessary conditions for deadlock, along with hold and wait, no preemption, and circular wait.

Q58

U4: Deadlocks

How many conditions must hold simultaneously for a deadlock to occur?

A Two

B Three

C Four

D Five

Think about your answer...

Q58 ANSWER

How many conditions must hold simultaneously for a deadlock to occur?

A Two

B Three

C **Four** ✓

D Five

EXPLANATION

All four conditions (mutual exclusion, hold and wait, no preemption, circular wait) must hold simultaneously for a deadlock to occur.

Q59

U4: Deadlocks

Deadlock prevention ensures that at least one of the necessary conditions:

A Is always satisfied

B Is never satisfied

C Is partially satisfied

D Is satisfied during execution

Think about your answer...

Q59 ANSWER

Deadlock prevention ensures that at least one of the necessary conditions:

A Is always satisfied

B **Is never satisfied** ✓

C Is partially satisfied

D Is satisfied during execution

EXPLANATION

Deadlock prevention ensures that at least one of the necessary conditions cannot hold, making deadlock impossible.

Q60

U4: Deadlocks

Banker's algorithm is used for deadlock:

A Prevention

B Avoidance

C Detection

D Recovery

Think about your answer...

Q60 ANSWER

Banker's algorithm is used for deadlock:

A Prevention

B **Avoidance** ✓

C Detection

D Recovery

EXPLANATION

Banker's algorithm is used for deadlock avoidance by checking if granting a request leads to a safe state.

Q61

U4: Deadlocks

In Banker's algorithm, a safe state means:

- **A** All processes are in deadlock
- **B** There exists a sequence in which all processes can complete
- **C** No process needs resources
- **D** The system is in an unsafe state

Think about your answer...

Q61 ANSWER

In Banker's algorithm, a safe state means:

A All processes are in deadlock

B **There exists a sequence in which all processes can complete** ✓

C No process needs resources

D The system is in an unsafe state

EXPLANATION

A safe state means there exists a sequence in which all processes can complete without causing deadlock.

Q62 U4: Deadlocks

The circular wait condition can be prevented by:

- **A** Allowing preemption
- **B** Imposing a total ordering of resource types
- **C** Using semaphores
- **D** Increasing the number of resources

Think about your answer...

Q62 ANSWER

The circular wait condition can be prevented by:

A Allowing preemption

B Imposing a total ordering of resource types ✓

C Using semaphores

D Increasing the number of resources

EXPLANATION

Circular wait can be prevented by imposing a total ordering on resource types and requiring processes to request resources in increasing order.

Deadlock detection algorithm uses:

- **A** Resource allocation graph
- **B** Page table
- **C** TLB
- **D** PCB

Think about your answer...

Q63 ANSWER

Deadlock detection algorithm uses:

A **Resource allocation graph** ✓

B Page table

C TLB

D PCB

EXPLANATION

The deadlock detection algorithm uses a resource allocation graph or wait-for graph to identify cycles indicating deadlock.

Which of the following is NOT a deadlock recovery technique?

- **A** Process termination
- **B** Resource preemption
- **C** Increasing time quantum
- **D** Rollback to a safe state

Think about your answer...

Q64

ANSWER

Which of the following is NOT a deadlock recovery technique?

A Process termination

B Resource preemption

C **Increasing time quantum** ✓

D Rollback to a safe state

EXPLANATION

Prevention is NOT a recovery technique; it is a proactive strategy. Recovery techniques include process termination and resource preemption.

Q65

U4: Deadlocks

A resource allocation graph with a cycle:

- **A** Always indicates a deadlock
- **B** Indicates a deadlock only if each resource has one instance
- **C** Never indicates a deadlock
- **D** Indicates starvation

Think about your answer...

Q65 ANSWER

A resource allocation graph with a cycle:

A Always indicates a deadlock

B **Indicates a deadlock only if each resource has one instance** ✓

C Never indicates a deadlock

D Indicates starvation

EXPLANATION

A cycle in a resource allocation graph indicates a potential deadlock (guaranteed deadlock if each resource has only one instance).

Q66 U4: Deadlocks

The hold and wait condition means a process:

- **A** Releases all resources before requesting new ones
- **B** Holds at least one resource and is waiting for additional resources
- **C** Never requests resources
- **D** Waits indefinitely without holding resources

Think about your answer...

Q66

ANSWER

The hold and wait condition means a process:

A Releases all resources before requesting new ones

B Holds at least one resource and is waiting for additional resources ✓

C Never requests resources

D Waits indefinitely without holding resources

EXPLANATION

The hold and wait condition means a process holds at least one resource while waiting for additional resources held by others.

Q67

U4: Deadlocks

In the Banker's algorithm, the Available array represents:

A Maximum resources needed by each process

B Currently available instances of each resource type

C Resources allocated to each process

D Total resources in the system

Think about your answer...

Q67 ANSWER

In the Banker's algorithm, the Available array represents:

A Maximum resources needed by each process

B Currently available instances of each resource type ✓

C Resources allocated to each process

D Total resources in the system

EXPLANATION

The Available array represents the number of available instances of each resource type that can be allocated to processes.

Q68

U4: Deadlocks

Which deadlock handling strategy ignores the problem entirely?

- **A** Prevention
- **B** Avoidance
- **C** Detection and recovery
- **D** Ostrich algorithm

Think about your answer...

Q68 ANSWER

Which deadlock handling strategy ignores the problem entirely?

A Prevention

B Avoidance

C Detection and recovery

D **Ostrich algorithm** ✓

EXPLANATION

The ostrich strategy (ignoring the problem) is used by most operating systems, assuming deadlocks occur rarely and the cost of prevention outweighs the benefit.

Q69

U4: Deadlocks

An unsafe state in the Banker's algorithm means:

- **A** Deadlock has definitely occurred
- **B** Deadlock may occur depending on process behavior
- **C** The system is running efficiently
- **D** All processes have completed

Think about your answer...

Q69 ANSWER

An unsafe state in the Banker's algorithm means:

- **A** Deadlock has definitely occurred
- **B** **Deadlock may occur depending on process behavior** ✓
- **C** The system is running efficiently
- **D** All processes have completed

EXPLANATION

An unsafe state means no sequence exists that can guarantee all processes will complete; deadlock MAY occur but is not guaranteed.

Q70

U4: Deadlocks

The no-preemption condition can be broken by:

- **A** Allowing processes to forcefully take resources from others
- **B** Using semaphores
- **C** Increasing memory
- **D** Using monitors

Think about your answer...

Q70 ANSWER

The no-preemption condition can be broken by:

A **Allowing processes to forcefully take resources from others** ✓

B Using semaphores

C Increasing memory

D Using monitors

EXPLANATION

The no-preemption condition can be broken by forcibly removing resources from a process and reallocating them.

Q71 U4: Deadlocks

In deadlock detection, if the system is found to be in a deadlock state, the common recovery method is:

- **A** Restarting the system
- **B** Terminating one or more processes involved in the deadlock
- **C** Adding more memory
- **D** Ignoring the deadlock

Think about your answer...

Q71

ANSWER

In deadlock detection, if the system is found to be in a deadlock state, the common recovery method is:

A Restarting the system

B Terminating one or more processes involved in the deadlock ✓

C Adding more memory

D Ignoring the deadlock

EXPLANATION

The common recovery method is process termination, either aborting all deadlocked processes or selectively aborting one at a time until the cycle is broken.

Unit 5

Memory Management

Questions 72 - 92

Q72

U5: Memory Management

Logical address is also known as:

- A Physical address
- B Virtual address
- C Absolute address
- D Real address

Think about your answer...

Q72 ANSWER

Logical address is also known as:

A Physical address

B **Virtual address** ✓

C Absolute address

D Real address

EXPLANATION

Logical address is also known as virtual address, generated by the CPU during program execution.

Q73

U5: Memory Management

Internal fragmentation occurs in:

A. Variable partitioning

B. Fixed partitioning

C. Paging

D. Both B and C

Think about your answer...

Q73 ANSWER

Internal fragmentation occurs in:

A Variable partitioning

B Fixed partitioning

C Paging

D Both B and C ✓

EXPLANATION

Internal fragmentation occurs in fixed/pure segmentation where allocated memory may be larger than needed, wasting space within a partition.

Q74

U5: Memory Management

External fragmentation occurs in:

- **A** Fixed partitioning
- **B** Variable partitioning
- **C** Paging
- **D** Segmentation with paging

Think about your answer...

Q74 ANSWER

External fragmentation occurs in:

- A Fixed partitioning
- B **Variable partitioning** ✓
- C Paging
- D Segmentation with paging

EXPLANATION

External fragmentation occurs in variable partition allocation where free memory exists but is scattered in small, non-contiguous blocks.

Q75

U5: Memory Management

Compaction is a solution for:

A Internal fragmentation

B External fragmentation

C Paging overhead

D Thrashing

Think about your answer...

Q75

ANSWER

Compaction is a solution for:

A Internal fragmentation

B **External fragmentation** ✓

C Paging overhead

D Thrashing

EXPLANATION

Compaction is a solution for external fragmentation, moving all free memory together to create one large contiguous block.

Q76

U5: Memory Management

In paging, the physical memory is divided into:

- A Segments of variable size
- B Frames of fixed size
- C Pages of variable size
- D Blocks of different sizes

Think about your answer...

Q76

ANSWER

In paging, the physical memory is divided into:

A Segments of variable size

B **Frames of fixed size** ✓

C Pages of variable size

D Blocks of different sizes

EXPLANATION

In paging, physical memory is divided into fixed-size blocks called frames.

Q77

U5: Memory Management

A page table is used to map:

- **A** Physical address to physical address
- **B** Logical address to physical address
- **C** Virtual address to logical address
- **D** Frame number to segment number

Think about your answer...

Q77 ANSWER

A page table is used to map:

A Physical address to physical address

B **Logical address to physical address** ✓

C Virtual address to logical address

D Frame number to segment number

EXPLANATION

A page table maps logical (virtual) page numbers to physical frame numbers, enabling address translation.

Q78

U5: Memory Management

A page fault occurs when:

A. The page is in main memory

B. The referenced page is not in main memory

C. The process terminates

D. The CPU is idle

Think about your answer...

Q78 ANSWER

A page fault occurs when:

A The page is in main memory

B The referenced page is not in main memory ✓

C The process terminates

D The CPU is idle

EXPLANATION

A page fault occurs when a process references a page that is not currently loaded in physical memory.

Q79

U5: Memory Management

The optimal page replacement algorithm replaces the page that:

- **A** Was most recently used
- **B** Will not be used for the longest period
- **C** Was least recently used
- **D** Was first brought into memory

Think about your answer...

Q79 ANSWER

The optimal page replacement algorithm replaces the page that:

A Was most recently used

B Will not be used for the longest period ✓

C Was least recently used

D Was first brought into memory

EXPLANATION

The optimal page replacement algorithm replaces the page that will not be used for the longest period in the future.

Q80

U5: Memory Management

The LRU page replacement algorithm replaces the page that:

A Was most recently used

B Has not been used for the longest time

C Will be used next

D Has the highest frequency of use

Think about your answer...

Q80

ANSWER

The LRU page replacement algorithm replaces the page that:

A Was most recently used

B Has not been used for the longest time ✓

C Will be used next

D Has the highest frequency of use

EXPLANATION

The LRU algorithm replaces the page that has not been used for the longest time in the past.

Thrashing occurs when:

- **A** A process spends more time executing than paging
- **B** A process spends more time paging than executing
- **C** There is no paging activity
- **D** The CPU utilization is very high

Think about your answer...

Q81 ANSWER

Thrashing occurs when:

A. A process spends more time executing than paging

B. A process spends more time paging than executing ✓

C. There is no paging activity

D. The CPU utilization is very high

EXPLANATION

Thrashing occurs when a process spends more time paging (swapping pages in and out) than executing, due to insufficient memory.

Q82

U5: Memory Management

The working set model is used to:

A Prevent deadlock

B Prevent thrashing

C Improve CPU scheduling

D Manage file systems

Think about your answer...

Q82 ANSWER

The working set model is used to:

A Prevent deadlock

B **Prevent thrashing** ✓

C Improve CPU scheduling

D Manage file systems

EXPLANATION

The working set model is used to determine the set of pages a process needs to keep in memory to avoid thrashing.

Q83 U5: Memory Management

In demand paging, a page is loaded into memory:

- **A** When the program starts
- **B** Only when it is demanded/referenced
- **C** Before the program starts
- **D** At compile time

Think about your answer...

Q83 **ANSWER**

In demand paging, a page is loaded into memory:

A When the program starts

B Only when it is demanded/referenced ✓

C Before the program starts

D At compile time

EXPLANATION

In demand paging, a page is loaded into memory only when it is referenced (on demand), not in advance.

Q84

U5: Memory Management

The dirty bit indicates:

A The page is corrupted

B The page has been modified since being loaded

C The page is read-only

D The page is shared

Think about your answer...

Q84

ANSWER

The dirty bit indicates:

A The page is corrupted

B **The page has been modified since being loaded** ✓

C The page is read-only

D The page is shared

EXPLANATION

The dirty bit indicates whether a page has been modified since being loaded into memory, determining if it needs to be written back to disk.

Q85

U5: Memory Management

The FIFO page replacement algorithm may suffer from:

A Belady's Anomaly

B Starvation

C Deadlock

D Aging

Think about your answer...

Q85 ANSWER

The FIFO page replacement algorithm may suffer from:

A Belady's Anomaly ✓

B Starvation

C Deadlock

D Aging

EXPLANATION

FIFO may suffer from Belady's Anomaly, where increasing the number of frames can result in more page faults.

Q86

U5: Memory Management

Segmentation divides the logical address space into:

A Fixed-size pages

B Variable-size segments

C Equal-sized frames

D Uniform blocks

Think about your answer...

Q86 ANSWER

Segmentation divides the logical address space into:

A Fixed-size pages

B Variable-size segments ✓

C Equal-sized frames

D Uniform blocks

EXPLANATION

Segmentation divides the logical address space into variable-sized segments based on the logical structure of the program.

Q87

U5: Memory Management

Virtual memory allows execution of processes that are:

- **A** Completely in physical memory
- **B** Not completely in physical memory
- **C** Only in swap space
- **D** Only in cache

Think about your answer...

Q87 ANSWER

Virtual memory allows execution of processes that are:

- **A** Completely in physical memory
- **B** **Not completely in physical memory** ✓
- **C** Only in swap space
- **D** Only in cache

EXPLANATION

Virtual memory allows execution of processes that are not completely in memory, enabling processes larger than physical memory.

Q88

U5: Memory Management

TLB (Translation Lookaside Buffer) is used to:

A Store pages

B Speed up page table lookups

C Manage disk I/O

D Handle interrupts

Think about your answer...

Q88 ANSWER

TLB (Translation Lookaside Buffer) is used to:

A Store pages

B **Speed up page table lookups** ✓

C Manage disk I/O

D Handle interrupts

EXPLANATION

TLB is a hardware cache used to speed up virtual-to-physical address translation by storing recently used page table entries.

Q89

U5: Memory Management

Locality of reference refers to:

- **A** The tendency of programs to access the same set of memory locations repeatedly
- **B** The number of pages in memory
- **C** The size of the page table
- **D** The number of segments in a program

Think about your answer...

Q89

ANSWER

Locality of reference refers to:

A The tendency of programs to access the same set of memory locations repeatedly ✓

B The number of pages in memory

C The size of the page table

D The number of segments in a program

EXPLANATION

Locality of reference refers to the tendency of programs to access a relatively small set of memory locations within a given time interval.

Q90

U5: Memory Management

Which page replacement algorithm gives the minimum page faults?

A FIFO

B LRU

C Optimal

D Second Chance

Think about your answer...

Q90 ANSWER

Which page replacement algorithm gives the minimum page faults?

A FIFO

B LRU

C **Optimal** ✓

D Second Chance

EXPLANATION

The optimal page replacement algorithm gives the minimum page faults, but it requires future knowledge and cannot be implemented in practice.

Q91

U5: Memory Management

In the Second Chance page replacement algorithm, a reference bit of:

A 0 means the page was recently used

B 1 means the page was recently used

C 0 means the page is dirty

D 1 means the page is in deadlock

Think about your answer...

Q91

ANSWER

In the Second Chance page replacement algorithm, a reference bit of:

A 0 means the page was recently used

B 1 means the page was recently used ✓

C 0 means the page is dirty

D 1 means the page is in deadlock

EXPLANATION

In the Second Chance algorithm, a page with reference bit 0 is replaced; a page with reference bit 1 gets a second chance (bit reset, moved to tail).

Q92

U5: Memory Management

The Not Recently Used (NRU) algorithm classifies pages based on:

- **A** Only the reference bit
- **B** Only the modify bit
- **C** Both reference and modify bits
- **D** Neither reference nor modify bits

Think about your answer...

Q92 ANSWER

The Not Recently Used (NRU) algorithm classifies pages based on:

A Only the reference bit

B Only the modify bit

C Both reference and modify bits ✓

D Neither reference nor modify bits

EXPLANATION

The NRU algorithm classifies pages based on reference and modify bits into four classes, replacing from the lowest non-empty class.

Unit 6

File Management

Questions 93 - 112

Q93 U6: File Management

A file is a collection of:

- **A** Random data
- **B** Related information stored on secondary storage
- **C** Only executable code
- **D** Only text data

Think about your answer...

Q93 ANSWER

A file is a collection of:

A Random data

B **Related information stored on secondary storage** ✓

C Only executable code

D Only text data

EXPLANATION

A file is a collection of related information (data or programs) defined by its creator, stored on secondary storage.

Q94

U6: File Management

Which file access method accesses records in sequential order?

A Direct access

B Sequential access

C Indexed access

D Random access

Think about your answer...

Q94

ANSWER

Which file access method accesses records in sequential order?

A Direct access

B Sequential access ✓

C Indexed access

D Random access

EXPLANATION

Sequential access method accesses records in a predetermined order, one after another, as in magnetic tape.

Q95 U6: File Management

In contiguous file allocation, each file occupies:

- **A** Scattered blocks across the disk
- **B** A set of contiguous blocks on the disk
- **C** Only one block
- **D** Blocks linked by pointers

Think about your answer...

Q95 ANSWER

In contiguous file allocation, each file occupies:

- **A** Scattered blocks across the disk
- **B** **A set of contiguous blocks on the disk ✓**
- **C** Only one block
- **D** Blocks linked by pointers

EXPLANATION

In contiguous allocation, each file occupies a set of contiguous blocks on disk, providing fast access but causing external fragmentation.

Linked allocation solves the problem of:

- **A** Internal fragmentation
- **B** External fragmentation
- **C** Both internal and external fragmentation
- **D** Thrashing

Think about your answer...

Q96 ANSWER

Linked allocation solves the problem of:

A Internal fragmentation

B **External fragmentation** ✓

C Both internal and external fragmentation

D Thrashing

EXPLANATION

Linked allocation solves the problem of external fragmentation by using pointers to link non-contiguous disk blocks.

Q97 U6: File Management

The indexed allocation method uses:

- **A** A linked list of blocks
- **B** An index block containing pointers to data blocks
- **C** Contiguous blocks only
- **D** A bit vector

Think about your answer...

Q97 ANSWER

The indexed allocation method uses:

- **A** A linked list of blocks
- **B** **An index block containing pointers to data blocks** ✓
- **C** Contiguous blocks only
- **D** A bit vector

EXPLANATION

Indexed allocation uses an index block that contains pointers to all the blocks of a file, allowing direct access.

Q98 U6: File Management

A directory structure that allows shared subdirectories and files is:

- **A** Single-level directory
- **B** Two-level directory
- **C** Tree-structured directory
- **D** Acyclic-graph directory

Think about your answer...

Q98

ANSWER

A directory structure that allows shared subdirectories and files is:

A Single-level directory

B Two-level directory

C Tree-structured directory

D **Acyclic-graph directory** ✓

EXPLANATION

An acyclic-graph directory structure allows shared subdirectories and files through multiple paths without creating cycles.

Q99

U6: File Management

Free-space management using a bit vector uses:

A One bit per disk block

B One byte per disk block

C One word per disk block

D A linked list of free blocks

Think about your answer...

Q99 ANSWER

Free-space management using a bit vector uses:

- **A. One bit per disk block** ✓
- B. One byte per disk block
- C. One word per disk block
- D. A linked list of free blocks

EXPLANATION

A bit vector (bitmap) uses one bit per block, where 0 indicates free and 1 indicates allocated, providing efficient free-space management.

Q100 U6: File Management

The FCFS disk scheduling algorithm processes requests in:

- **A** The order of shortest seek time
- **B** The order they arrive
- **C** The reverse order
- **D** Random order

Think about your answer...

Q100

ANSWER

The FCFS disk scheduling algorithm processes requests in:

A The order of shortest seek time

B The order they arrive ✓

C The reverse order

D Random order

EXPLANATION

FCFS disk scheduling processes requests in the order they arrive, with no optimization for seek time.

Q101

U6: File Management

The SSTF disk scheduling algorithm selects the request:

- **A** That arrived first
- **B** Closest to the current head position
- **C** Farthest from the current head position
- **D** With the highest priority

Think about your answer...

Q101

ANSWER

The SSTF disk scheduling algorithm selects the request:

A That arrived first

B Closest to the current head position ✓

C Farthest from the current head position

D With the highest priority

EXPLANATION

SSTF (Shortest Seek Time First) selects the request closest to the current head position to minimize seek time.

Q102 U6: File Management

In the SCAN disk scheduling algorithm, the disk arm moves:

- **A** Only in one direction
- **B** In one direction then reverses at the end
- **C** Randomly
- **D** To the center only

Think about your answer...

Q102 ANSWER

In the SCAN disk scheduling algorithm, the disk arm moves:

- **A** Only in one direction
- **B** **In one direction then reverses at the end** ✓
- **C** Randomly
- **D** To the center only

EXPLANATION

In SCAN, the disk arm moves in one direction servicing requests until reaching the end, then reverses direction.

Q103

U6: File Management

C-SCAN (Circular SCAN) improves upon SCAN by:

A Moving in both directions servicing requests

B Moving in only one direction and quickly returning to the start

C Not moving at all

D Always moving to the center

Think about your answer...

Q103 ANSWER

C-SCAN (Circular SCAN) improves upon SCAN by:

A Moving in both directions servicing requests

B **Moving in only one direction and quickly returning to the start** ✓

C Not moving at all

D Always moving to the center

EXPLANATION

C-SCAN improves upon SCAN by treating the disk as circular, moving only in one direction and jumping back to the beginning, providing more uniform wait times.

Q104 U6: File Management

DMA (Direct Memory Access) allows:

- **A** CPU to transfer data to I/O devices
- **B** I/O devices to transfer data directly to/from memory
- **C** Only memory-to-memory transfer
- **D** Only CPU-to-CPU transfer

Think about your answer...

Q104 ANSWER

DMA (Direct Memory Access) allows:

A CPU to transfer data to I/O devices

B I/O devices to transfer data directly to/from memory ✓

C Only memory-to-memory transfer

D Only CPU-to-CPU transfer

EXPLANATION

DMA allows peripherals to transfer data directly to/from memory without CPU involvement, reducing CPU overhead.

Q105 U6: File Management

A device driver is:

- **A** A hardware component
- **B** Software that provides an interface for the OS to communicate with a device
- **C** A type of file system
- **D** A memory management technique

Think about your answer...

Q105 ANSWER

A device driver is:

A. A hardware component

B. **Software that provides an interface for the OS to communicate with a device** ✓

C. A type of file system

D. A memory management technique

EXPLANATION

A device driver is a software routine that provides an interface between the OS and hardware devices, translating OS commands to device-specific operations.

Q106 U6: File Management

The boot block contains:

- **A** User data
- **B** The initial program that loads the operating system
- **C** File allocation table
- **D** Directory structure

Think about your answer...

Q106 ANSWER

The boot block contains:

A User data

B **The initial program that loads the operating system** ✓

C File allocation table

D Directory structure

EXPLANATION

The boot block contains the bootstrap program (bootloader) that initializes the OS when the computer starts up.

Q107

U6: File Management

Bad blocks on a disk are handled by:

A Ignoring them

B Marking them and using spare blocks or skipping them

C Reformatting the entire disk

D Restarting the system

Think about your answer...

Q107 ANSWER

Bad blocks on a disk are handled by:

A Ignoring them

B Marking them and using spare blocks or skipping them ✓

C Reformatting the entire disk

D Restarting the system

EXPLANATION

Bad blocks on a disk are handled by the disk controller, which substitutes spare blocks and maintains a bad-block list.

Disk formatting involves:

- **A** Only deleting files
- **B** Dividing the disk into sectors for the OS to use
- **C** Only checking for bad blocks
- **D** Only creating directories

Think about your answer...

Q108 ANSWER

Disk formatting involves:

A Only deleting files

B **Dividing the disk into sectors for the OS to use** ✓

C Only checking for bad blocks

D Only creating directories

EXPLANATION

Disk formatting involves dividing the disk into sectors so the disk controller can read and write data, including low-level and high-level formatting.

Q109

U6: File Management

A hash table in directory implementation provides:

A Slower access than linear list

B Faster search time for file names

C Sequential access only

D No collision handling

Think about your answer...

Q109 ANSWER

A hash table in directory implementation provides:

A Slower access than linear list

B **Faster search time for file names** ✓

C Sequential access only

D No collision handling

EXPLANATION

A hash table in directory implementation provides fast O(1) average-case lookup time for file name resolution.

Q110

U6: File Management

Device-independent I/O software provides:

- **A** Hardware-specific functions
- **B** A uniform interface to device drivers for user programs
- **C** Direct hardware access
- **D** Memory management

Think about your answer...

Q110

ANSWER

Device-independent I/O software provides:

A Hardware-specific functions

B A uniform interface to device drivers for user programs ✓

C Direct hardware access

D Memory management

EXPLANATION

Device-independent I/O software provides a uniform interface to device drivers, allowing applications to access devices without knowing hardware details.

Q111

U6: File Management

The primary goal of I/O software is:

- **A** To maximize hardware cost
- **B** To provide a uniform, device-independent interface
- **C** To slow down I/O operations
- **D** To eliminate device drivers

Think about your answer...

Q111

ANSWER

The primary goal of I/O software is:

A To maximize hardware cost

B To provide a uniform, device-independent interface ✓

C To slow down I/O operations

D To eliminate device drivers

EXPLANATION

The primary goal of I/O software is to provide a uniform, device-independent interface that hides hardware details from applications.

Q112 U6: File Management

In the linked list method of free-space management, each free block contains:

- **A** Data
- **B** A pointer to the next free block
- **C** An index
- **D** A bit map

Think about your answer...

Q112

ANSWER

In the linked list method of free-space management, each free block contains:

A Data

B A pointer to the next free block ✓

C An index

D A bit map

EXPLANATION

In the linked list method, each free block contains a pointer to the next free block, forming a chain of free disk blocks.